

EXPNO:3	CLASSIFICATION WITH DECISION TREES
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AIM

To implement a Decision Tree classifier and evaluate its performance using **accuracy score** and **confusion matrix** on a real-world dataset.

ALGORITHM

1. Import necessary libraries
2. Load a classification dataset (e.g., Iris or Titanic)
3. Split the dataset into training and test sets
4. Preprocess data if needed
5. Train a DecisionTreeClassifier from sklearn.tree
6. Predict on test data
7. Evaluate using:
 - Confusion Matrix
 - Accuracy Score
8. Visualize the Decision Tree (optional)

CODE:

```
import pandas as pd

import matplotlib.pyplot as plt

from google.colab import files

from sklearn.preprocessing import LabelEncoder

from sklearn.tree import DecisionTreeClassifier, plot_tree

from sklearn.metrics import confusion_matrix, accuracy_score, ConfusionMatrixDisplay

uploaded = files.upload()

df = pd.read_csv("college_majors_2026.csv")

print("Dataset Shape :", df.shape)

print("\nFirst Five Rows")
```

```
print(df.head())

df = df.dropna()

target = "credential_name"

encoders = {}

for col in df.columns:

    if df[col].dtype == "object":

        le = LabelEncoder()

        df[col] = le.fit_transform(df[col].astype(str))

        encoders[col] = le


X = df.drop(columns=[target])

y = df[target]

model = DecisionTreeClassifier(random_state=42)

model.fit(X, y)

y_pred = model.predict(X)

accuracy = accuracy_score(y, y_pred)

print("\nAccuracy Score =", accuracy * 100, "%")

cm = confusion_matrix(y, y_pred)

plt.figure(figsize=(10, 8))

disp = ConfusionMatrixDisplay(confusion_matrix=cm)

disp.plot(cmap="Blues", values_format="d")

plt.title("Confusion Matrix")

plt.show()

plt.figure(figsize=(25, 15))

plot_tree(

    model,

    feature_names=X.columns,

    filled=True,

    rounded=True,

    fontsize=7,
```

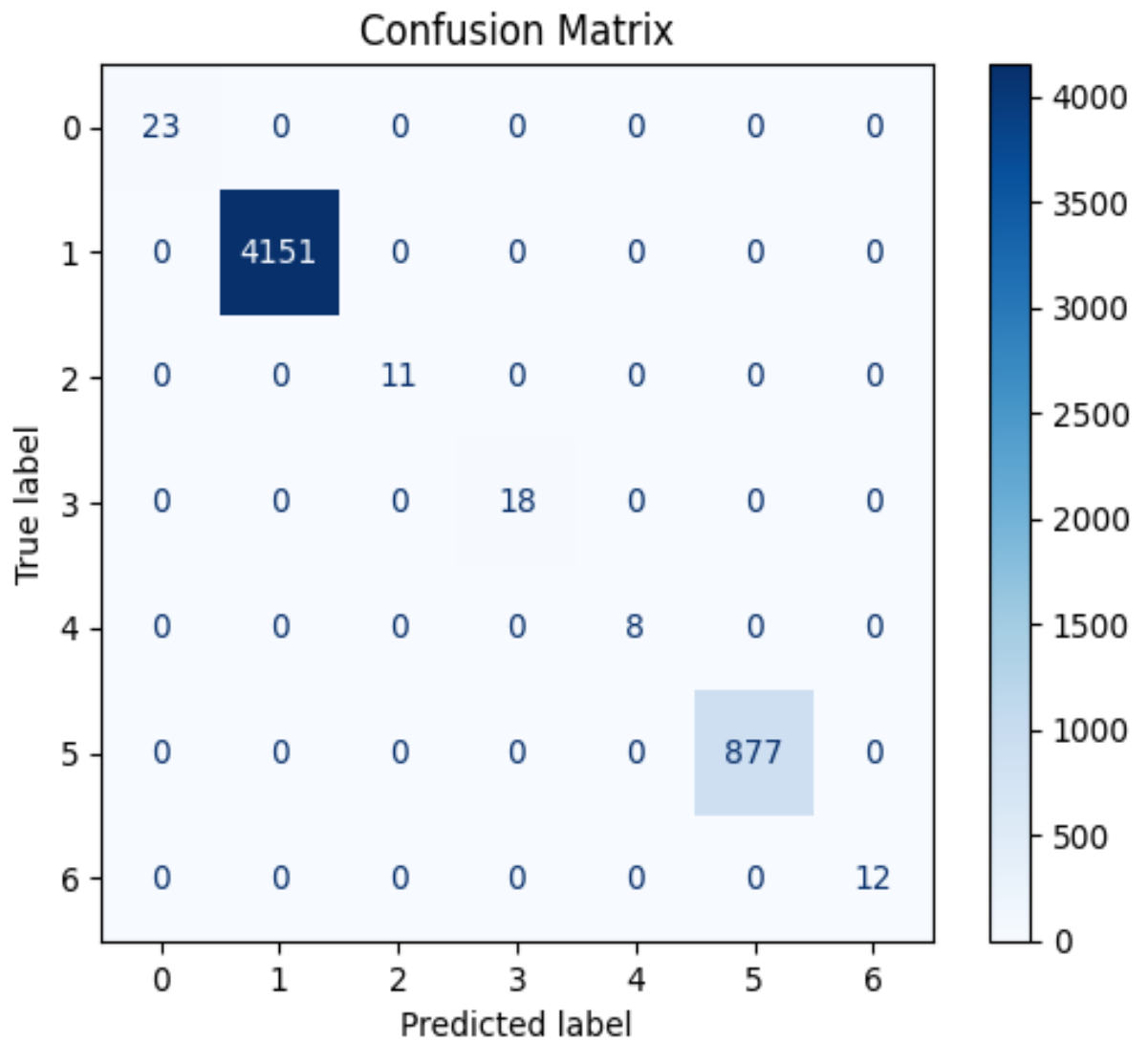
```
max_depth=3
)
plt.title("Decision Tree Visualization")
plt.show()
```

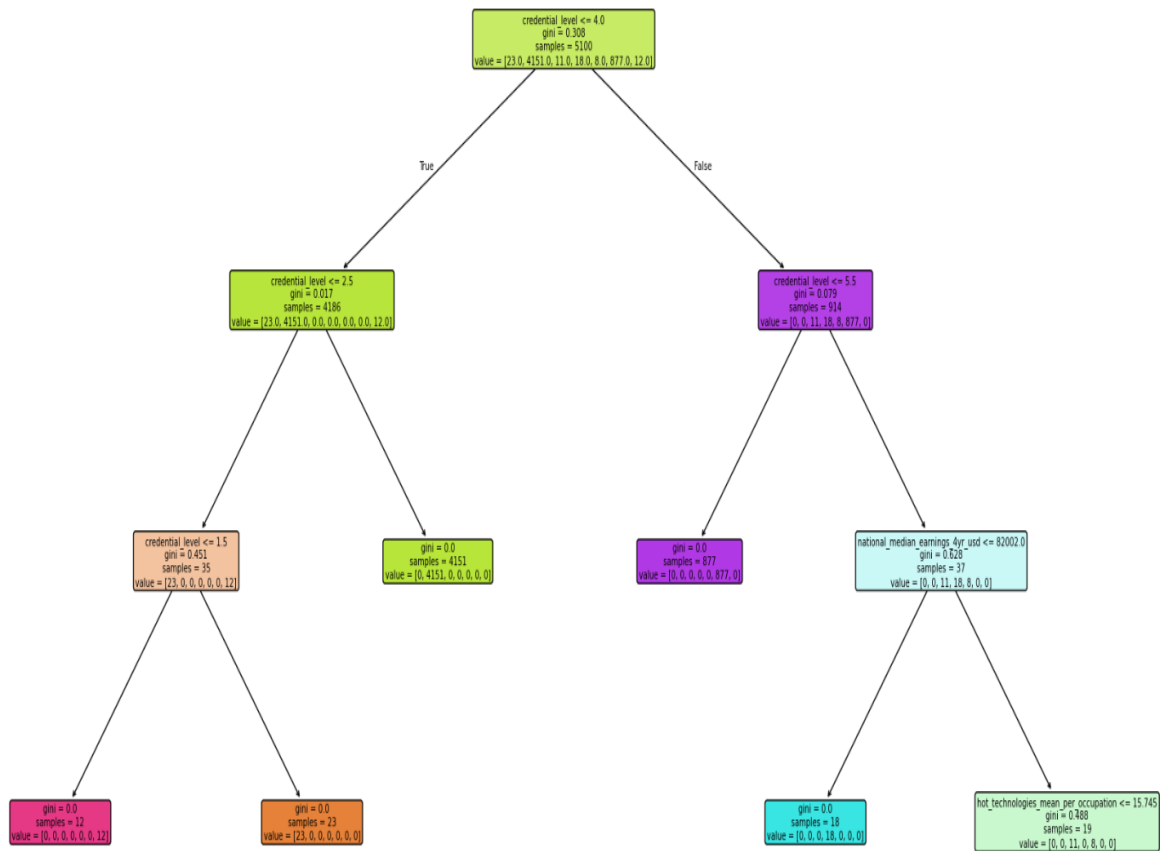
OUTPUT:

```
[5 rows x 72 columns]
```

```
Accuracy Score = 100.0 %
```

```
<Figure size 1000x800 with 0 Axes>
```





RESULT:

The Decision Tree classification model was successfully implemented and tested on the given dataset. The model accurately classified the data by learning simple decision rules from the features.

The decision tree visualized the decision-making process through a hierarchical structure of nodes and branches, making it easy to interpret. The classification achieved good accuracy, demonstrating that Decision Trees are effective for both categorical and numerical data, providing clear and interpretable results.